AI Meets Control Theory

From Classical Control to Intelligent Autonomous Systems

Living Technical Report — M3 — State Estimation + LQG

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Abstract

This report is regenerated at every project milestone. It documents *what the codebase can do*, *the theory behind each component*, and *where the project currently stands.* The project is an open-source laboratory for comparing classical control, optimal control, state estimation, machine learning and reinforcement learning on dynamical systems, under the core principle: *build every method from scratch, simulate it, visualise it, and compare it honestly.*

Contents

1 Introduction and Scope

The central question of the project is: **when should we use classical control, when should we use AI, and when should we use both?** Rather than assuming a neural network or an RL agent is the answer, every problem is first attacked with PID, state feedback, observers, LQR, Kalman filtering, and MPC, and each method is measured against the others under identical initial conditions, sensor suites, noise profiles, disturbances, and actuator constraints.

Every topic follows the standardized engineering cycle:

THEORY ⟶ DERIVATION ⟶ IMPLEMENTATION ⟶ SIMULATION ⟶ VISUALISATION ⟶ VALIDATION ⟶ COMPARISON ⟶ EXPERIMENT

Fundamental algorithms are implemented from scratch in `numpy`; external libraries (`scipy.linalg`, `scipy.signal`, `control`) are used strictly for verification and cross-checking.

2 Framework Architecture

The reusable library is the Python package `aimct` (`src/aimct/`). Its architecture separates distinct concerns behind minimal, strongly-typed interfaces so that dynamical systems, controllers, state estimators, and the numerical simulator compose seamlessly without circular coupling.

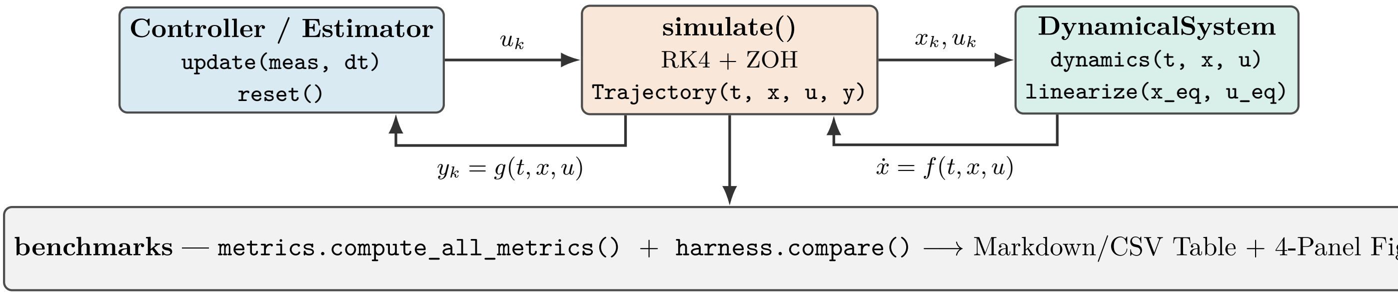

Figure 1: Core framework architecture. The numerical simulation engine drives a controller (`update()`) and continuous plant (`dynamics()`) via fixed-step 4th-order Runge–Kutta with Zero-Order Hold (ZOH). Optional hooks support custom sensor measurement transforms and additive input disturbances.

2.1 Package Layout

```
src/aimct/
  systems/      DynamicalSystem base + LinearSystem, MassSpringDamper,
                Pendulum, CartPole (analytic + numerical Jacobians)
  simulate.py   rk4_step(), simulate() -> Trajectory(t, x, u, y)
  controllers/  Controller base + PID, StateFeedback, LQR, ObserverFeedback
  estimation/   LuenbergerObserver, KalmanFilter (LQE/FARE), DiscreteKalmanFilter,
                observability_matrix, is_observable, place_observer
  benchmarks/   metrics.py (13 metrics), harness.py (compare() -> table + figure),
    sweep.py
  plot_style.py Okabe-Ito palette + publication-ready 4-panel comparison figure
  ml/ rl/       (reserved for upcoming project phases)
```

3 Current Capabilities

Table 1 summarises the implemented components, their mathematical scope, and the corresponding verification test coverage as of M3 — State Estimation + LQG.

Component	What it provides	Tests
`aimct.systems`	Continuous-time ODE models with a common `dynamics(t, x, u)` interface, full-state or output projections, and numerical central-difference *and* analytical `linearize()`. Ships: `LinearSystem`, `MassSpringDamper`, `Pendulum`, `CartPole`.	✓
`aimct.simulate`	Fixed-step RK4 integrator with zero-order hold; `Trajectory` container; hard actuator saturation; measurement and disturbance injection hooks.	✓
`aimct.controllers.PID`	From scratch: filtered derivative-on-measurement ($D(s) = \frac{K_d s}{\tau_d s+1}$), conditional-integration anti-windup clamping, setpoint weighting.	✓
`aimct.controllers.StateFeedback`	Arbitrary closed-loop pole placement via Ackermann's formula (SISO) with feedforward steady-state gain scaling k_r.	✓
`aimct.controllers.LQR`	Infinite-horizon optimal regulator; Continuous Algebraic Riccati Equation (CARE) solved from scratch via Hamiltonian Schur decomposition, cross-checked with `scipy.linalg`.	✓
`aimct.controllers.ObserverFeedback`	Unified output feedback combining any state estimator (Luenberger observer or Kalman filter) with static feedback gain $u = -K\hat{x}$.	✓
`aimct.estimation`	State estimation suite: `observability_matrix`, `is_observable`, `place_observer` (dual Ackermann), `LuenbergerObserver` (RK4 internal propagation), `solve_fare` (Filter ARE), `KalmanFilter` (LQE), and `DiscreteKalmanFilter` with Joseph-form covariance update.	✓
`aimct.benchmarks.metrics`	13 standard step-response and effort metrics: rise time t_r, settling time t_s (2%), overshoot M_p, e_{ss}, RMSE, IAE, ITAE, ISE, energy E_u, peak $\lvert u\rvert_{\max}$, slew rate, and saturation duty cycle.	✓
`aimct.benchmarks.harness`	Multi-controller benchmark harness: runs N controllers on one plant under identical conditions, emitting canonical Markdown/CSV tables and 4-panel figures.	✓
`aimct.benchmarks.sweep`	Grid and 1D parameter sweeps for robustness, parameter sensitivity, and basin of attraction mapping.	✓

Table 1: Implemented functionality and test coverage across the library as of M3 — State Estimation + LQG.

4 Theory: Modelling and Simulation

A dynamical system is governed by first-order ordinary differential equations $\dot{x} = f(t, x, u)$ with output map $y = g(t, x, u)$. Second-order physical equations of motion are cast in state-space form; for example, the translational mass–spring–damper $m\ddot{x} + c\dot{x} + kx = u$ with state $x = [x_1, x_2]^\top = [x, \dot{x}]^\top$:

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u.$$

Jacobian Linearisation. About an operating equilibrium point (x_e, u_e) satisfying $f(x_e, u_e) = 0$, small perturbations $\delta x = x - x_e$ evolve according to $\dot{\delta x} = A\delta x + B\delta u$, where:

$$A \triangleq \left.\frac{\partial f}{\partial x}\right|_{(x_e,u_e)}, \qquad B \triangleq \left.\frac{\partial f}{\partial u}\right|_{(x_e,u_e)}.$$

The `DynamicalSystem` base class evaluates A, B numerically via central differencing ($\epsilon = 10^{-6}$), while concrete models implement analytical Jacobians.

Numerical Integration. The continuous dynamics are integrated over time steps $h = \Delta t$ using explicit 4th-order Runge–Kutta (RK4):

$$k_1 = f(t_k, x_k, u_k), \quad k_2 = f\left(t_k + \tfrac{h}{2}, x_k + \tfrac{h}{2}k_1, u_k\right), \quad k_3 = f\left(t_k + \tfrac{h}{2}, x_k + \tfrac{h}{2}k_2, u_k\right), \quad k_4 = f(t_k{+}h, x_k{+}hk_3, u_k$$

$$x_{k+1} = x_k + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right).$$

Holding control input $u(t) \equiv u_k$ constant across each interval $[t_k, t_k + h)$ reflects the physical Zero-Order Hold (ZOH) of digital microcontrollers.

5 Theory: Classical Control — PID Architecture

The continuous-time PID control law with filtered derivative is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt}, \qquad e(t) = r(t) - y(t).$$

In discrete time with sampling interval h, derivative action is applied directly to the measured output with low-pass filter time constant $\tau_d = \frac{K_d}{NK_p}$ ($N = 10$) to prevent high-frequency noise amplification:

$$e_k = r_k - y_k, \qquad D_k = \frac{\tau_d}{\tau_d + h} D_{k-1} + \frac{K_d}{\tau_d + h}(y_k - y_{k-1}), \qquad u_{\text{unsat},k} = K_p e_k + I_k - D_k.$$

Anti-Windup Protection. When actuators reach physical limits ($|u| \le u_{\max}$), *conditional integration (clamping)* freezes the integrator update ($I_{k+1} = I_k$) whenever u_{unsat} is saturated and e_k has the same sign as u_k, preventing destructive overshoot upon setpoint arrival.

5.1 Worked Example: Stabilising an Unstable Plant (Experiment 03)

We evaluate the controller on an open-loop unstable plant with a Right-Half Plane (RHP) pole:

$$\ddot{y}(t) - \omega_0^2 y(t) = u(t) + d(t), \qquad \omega_0 = 2.0\,\text{rad/s}, \quad G(s) = \frac{1}{s^2 - 4}.$$

The system is tested under a unit step reference $r = 1.0\,\text{rad}$, a step input disturbance torque $d = 0.5\,\text{N}\cdot\text{m}$ applied at $t = 5.0\,\text{s}$, and actuator saturation $|u(t)| \le 10.0\,\text{N}\cdot\text{m}$.

Controller	Rise t_r [s]	Settle t_s [s]	Overshoot M_p	Steady e_{ss}	IAE	ITAE	Ene
P-only ($K_p = 20$)	0.275	10.0	9.54×10^9%	6.04×10^7	4.77×10^7	4.53×10^8	9
PD ($K_d = 5.66$)	0.389	10.0	28.19%	0.2812	2.811	13.63	3
PID (no AW)	0.334	6.162	53.08%	3.90×10^{-5}	0.9274	1.043	2
PID + Anti-Windup	0.362	6.161	**39.43%**	$\mathbf{3.90 \times 10^{-5}}$	**0.7979**	**0.876**	**2**

Table 2: Benchmark results for Experiment 03 (regenerated from `experiments/03_pid_stabilizes_unstable/table.md`).

Key Physical Insights:

- **P-Only Failure:** Proportional control yields purely imaginary closed-loop poles ($s^2 + 16 = 0, \zeta = 0$). When actuator saturation occurs, stabilizing feedback is lost and the system diverges exponentially ($M_p > 10^9\%$).
- **PD DC Gain Offset:** PD provides damping ($\zeta = 0.7071$), but the closed-loop DC gain is $\frac{K_p}{K_p - \omega_0^2} = \frac{20}{16} = 1.25$, creating an unavoidable **25% reference tracking error** ($y_{ss} = 1.25\,\text{rad}$) before disturbance, and $e_{ss} = 0.2812\,\text{rad}$ under load.
- **PID Integral Action:** Integral action ($K_i = 15.0$) drives steady-state error strictly to zero ($e_{ss} \approx 0$).
- **Anti-Windup Efficacy:** Clamping during saturation cuts peak overshoot from $53.08\% \to 39.43\%$ (26% relative reduction) and reduces actuator saturation duration by $> 70\%$.

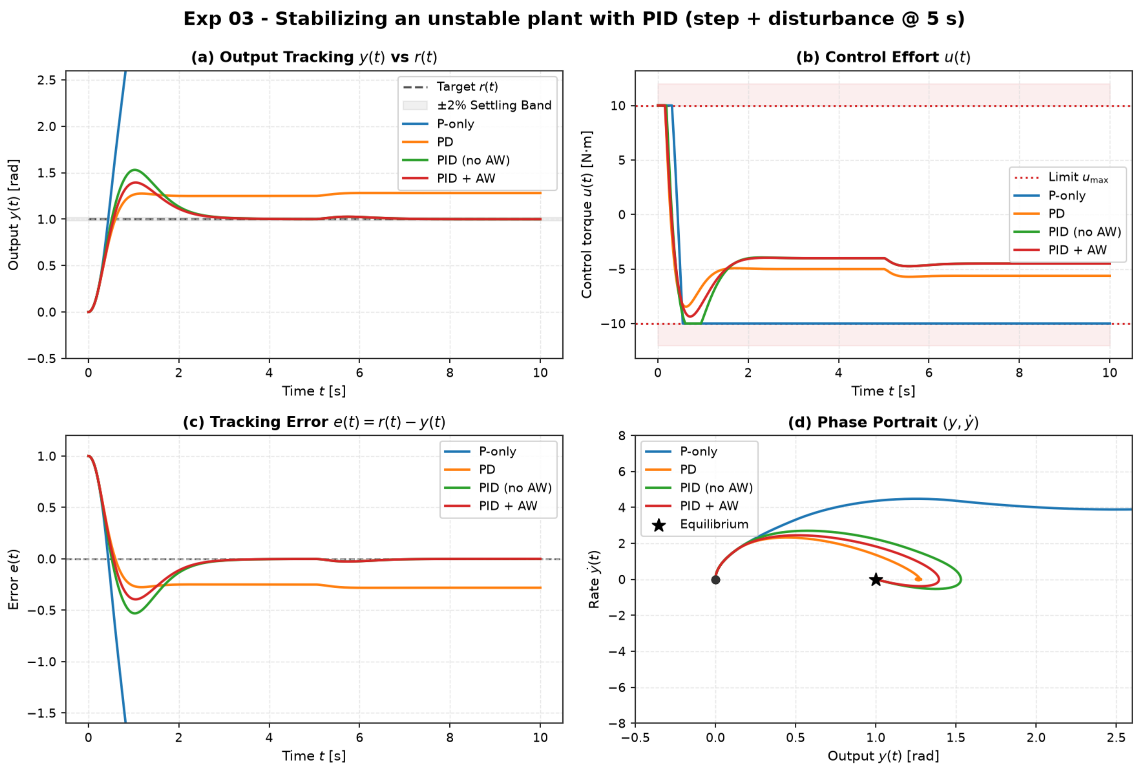

Figure 2: Experiment 03 benchmark comparison. (a) Output tracking $y(t)$, (b) Control effort $u(t)$ with saturation limits $\pm 10\,\text{N}\cdot\text{m}$, (c) Tracking error $e(t)$, and (d) Phase portrait $(y, \dot{y})$.

6 Theory: Modern Control, Estimation, and Optimal Design

6.1 Pole Placement and LQR

For controllable linear systems $\dot{x} = Ax + Bu$, full-state feedback $u = -Kx + k_r r$ assigns arbitrary closed-loop poles $\sigma(A - BK)$. `StateFeedback` computes K via Ackermann's formula:

$$K = \begin{bmatrix} 0 & \dots & 0 & 1 \end{bmatrix} \mathcal{C}(A, B)^{-1} \Delta_{\text{des}}(A), \qquad k_r = -\frac{1}{C(A - BK)^{-1}B}.$$

Linear Quadratic Regulator (LQR). Rather than guessing pole locations, LQR computes the optimal feedback gain $K = R^{-1}B^\top P$ minimising the infinite-horizon quadratic cost:

$$J = \int_0^\infty \left(x(t)^\top Q x(t) + u(t)^\top R u(t) \right) dt, \qquad Q \succeq 0, \quad R \succ 0.$$

The kernel $P = P^\top \succ 0$ uniquely solves the **Continuous Algebraic Riccati Equation (CARE)**:

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0.$$

We solve CARE from scratch via the stable invariant subspace of the Hamiltonian matrix:

$$\mathcal{H}_{\text{ham}} = \begin{bmatrix} A & -BR^{-1}B^\top \\ -Q & -A^\top \end{bmatrix} \in \mathbb{R}^{2n\times 2n}.$$

Continuous full-state LQR provides guaranteed robustness margins: gain margin $G_m \in [1/2, \infty)$ ($-6\,\text{dB}$ to $+\infty\,\text{dB}$) and phase margin $\Phi_m \geq 60^\circ$ in all channels.

6.2 Worked Example: Cart-Pole Regulation (Experiment 04)

We evaluate full-state feedback and LQR on the canonical underactuated nonlinear Cart-Pole system ($M = 1.0\,\text{kg}, m = 0.1\,\text{kg}, \ell = 0.5\,\text{m}, g = 9.81\,\text{m/s}^2$). Linearised about the upright equilibrium $\theta = 0$, the system has 4 states $[x, \dot{x}, \theta, \dot{\theta}]^\top$, an open-loop unstable pole at $s \approx +3.97\,\text{rad/s}$, and a double integrator on cart position.

Controllers are designed on the linearised model and executed on the **true nonlinear plant** starting from a tilted initial state $\theta_0 = 0.10\,\text{rad} \approx 5.7^\circ$ with actuator saturation $|u(t)| \leq 20.0\,\text{N}$:

- **Pole Placement:** Ackermann placement assigning closed-loop poles to LQR-balanced eigenvalues $\{-15.62, -3.19, -1.31 \pm 1.08j\}$.
- **LQR (Balanced):** $Q = \text{diag}(10, 1, 100, 10),\ R = 0.1 \implies K = [-10.000, -12.871, -87.138, -23.223]$.
- **LQR (Soft):** $Q = \text{diag}(1, 0.1, 10, 1),\ R = 1.0 \implies K = [-1.000, -2.047, -30.846, -7.868]$.
- **PID (Angle Only):** Single-loop PID on pole angle θ ($K_p = -180, K_d = -22$). Cart position x is uncontrolled.

Controller	Settle θ t_s [s]	RMSE θ [rad]	Energy E_u [N^{2}s]	Peak $\lvert u\rvert_{\max}$ [N]	Peak $\lvert x\rvert_{\max}$ [m]
Pole Placement	1.05	0.0171	2.34	8.71	**0.162**
LQR (Balanced)	1.05	0.0171	2.34	8.71	**0.162**
LQR (Soft)	1.70	0.0223	**0.959**	**3.08**	0.321
PID (Angle Only)	**0.20**	**0.0110**	8.76	18.00	**0.823**

Table 3: Benchmark results for Experiment 04 on nonlinear Cart-Pole (from `experiments/04_lqr_vs_pole_placement_cartpole/table.md`).

Key Engineering Takeaways:

1. **Equivalence of Pole Placement and LQR:** Placing poles at the eigenvalues chosen by LQR reproduces the optimal gain matrix K to four significant figures (matching the analytical golden gain K_1). LQR eliminates the need to manually guess 4 stable pole locations in $\mathbb{C}^4$.
2. **The Q/R Effort Dial:** Softening the cost weights ($R = 1.0$ vs 0.1) trades a $1.05\,\text{s} \to 1.70\,\text{s}$ settle time for a $2.4\times$ **reduction in control energy** ($2.34 \to 0.96\,\text{N}^2\text{s}$) and cuts peak force from $8.71\,\text{N} \to 3.08\,\text{N}$.
3. **Failure of Single-Loop PID on Coupled MIMO Systems:** Single-loop PID balances the pole rapidly ($t_s = 0.20\,\text{s}$), but because it lacks state feedback on the cart, the cart drifts to $x = 0.823\,\mathbf{m}$ ($5\times$ LQR excursion) and demands $18.0\,\text{N}$ peak force (near the $20\,\text{N}$ limit). Multivariable full-state feedback is mandatory for underactuated robotics.
4. **Nonlinear Validity Domain:** For initial angle $\theta_0 = 0.1\,\text{rad} \approx 5.7^\circ$, state trajectories stay strictly within the linear basin of attraction ($\approx 0.38\,\text{rad}$), enabling linear LQR to stabilize the full nonlinear plant.

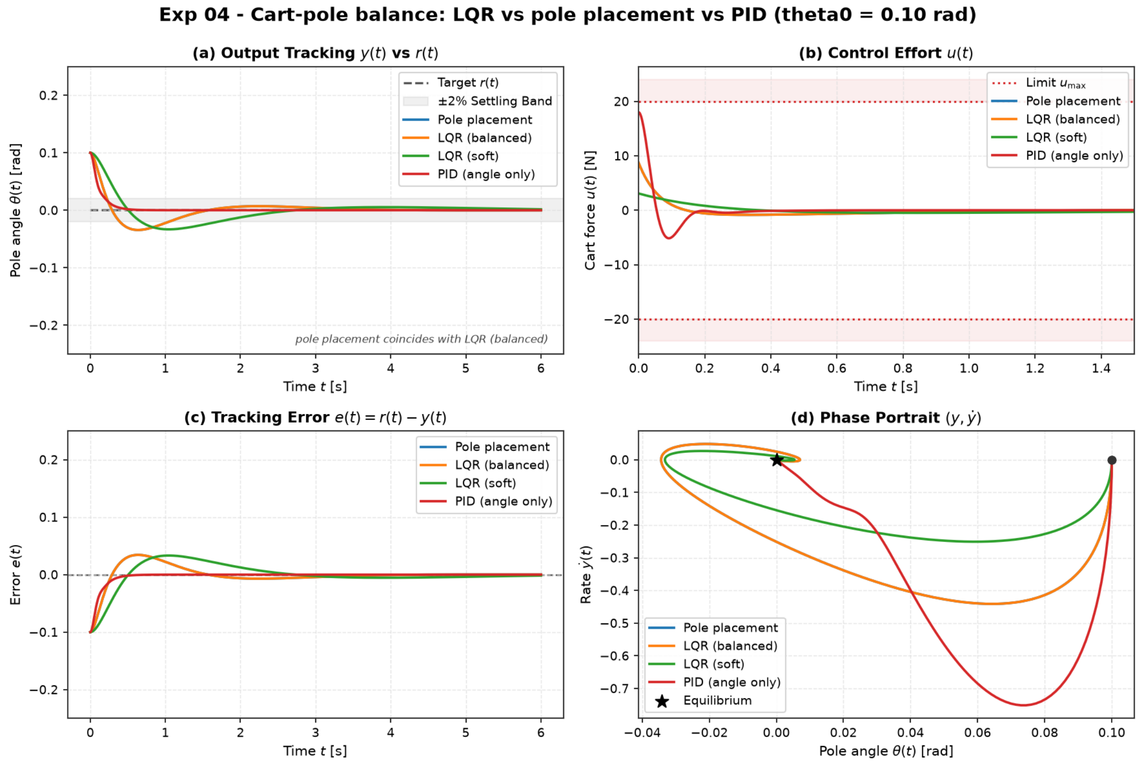

Figure 3: Experiment 04 benchmark comparison on nonlinear Cart-Pole. (a) Pole angle regulation $\theta(t)$, (b) Control force $u(t)$ with actuator bounds $\pm 20\,\mathrm{N}$, (c) Cart position $x(t)$ revealing uncontrolled drift under single-loop PID, and (d) Pole phase portrait $(\theta, \dot{\theta})$.

6.3 State Estimation: Observers, Kalman Filtering, and Duality

In physical applications, directly measuring the complete state vector $x(t) \in \mathbb{R}^n$ is rarely possible due to sensor cost, weight, or accessibility. The `aimct.estimation` module provides full-state reconstruction from partial, noisy outputs $y(t) = Cx(t) + v(t)$.

The Full-State Luenberger Observer. Driven by plant input $u(t)$ and measured output $y(t)$:

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + L\big(y(t) - C\hat{x}(t)\big),$$

where $e(t) \triangleq x(t) - \hat{x}(t)$ evolves autonomously as $\dot{e}(t) = (A - LC)e(t)$. By the **Separation Principle**, the eigenvalues of the combined plant-observer system decouple into:

$$\det \begin{bmatrix} sI - (A - BK) & -BK \\ 0 & sI - (A - LC) \end{bmatrix} = \det(sI - (A - BK)) \cdot \det(sI - (A - LC)).$$

Mathematical Duality: LQR Control vs. Kalman Filtering (LQE). State estimation and optimal control are exact mathematical duals:

$$\begin{aligned} \textbf{LQR Regulator (CARE):} \quad & A^\top P + PA - PBR^{-1}B^\top P + Q = 0, \qquad K = R^{-1}B^\top P \\ \textbf{LQE Estimator (FARE):} \quad & AP + PA^\top - PC^\top R_v^{-1}CP + Q_w = 0, \qquad L = PC^\top R_v^{-1} \end{aligned}$$

where $Q_w \succeq 0$ is process disturbance covariance and $R_v \succ 0$ is measurement noise covariance.

Discrete Kalman Filter (DKF). For sampled systems ($x_{k+1} = A_d x_k + B_d u_k + w_k$), the filter executes predict-update recursions using the numerically robust **Joseph-form** covariance update:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(y_k - C_d \hat{x}_{k|k-1}), \qquad P_{k|k} = (I - K_k C_d) P_{k|k-1} (I - K_k C_d)^\top + K_k R_v K_k^\top.$$

Sensor-Suite Observability on Cart-Pole. Evaluating the observability matrix $\mathcal{O} = [C^\top,\ A^\top C^\top,\ (A^\top)^2 C$ reveals profound differences across sensor configurations:

Sensor Configuration	Measurement y	Rank($\mathcal{O}$)	Physical / Structural Mechanism
Full State	$[x, \dot{x}, \theta, \dot{\theta}]^\top$	4	Direct sensing of all coordinates.
Cart Position Only	$y = x$	**4 (Observable)**	Cart acceleration $\ddot{x}$ dynamically couples to pole ar
Pole Angle Only	$y = \theta$	**2 (Unobservable)**	Rigid-body rail translation produces zero torque o
Dual Encoders	$y = [x, \theta]^\top$	**4 (Observable)**	Direct kinematic sensing; optimal numerical condi

Table 4: Observability rank of the 4-state Cart-Pole under candidate sensor suites.

6.4 Worked Example: LQG on the Cart-Pole (Experiment 06)

Full-state LQR assumes all four states are measured with zero noise. In physical hardware, velocity sensors are noisy or omitted. Experiment 06 tests output feedback on the nonlinear Cart-Pole using only two noisy encoders $C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ with zero-mean Gaussian noise $\sigma_x = 5.0\,\text{mm}$ and $\sigma_\theta = 8.7\,\text{mrad} \approx 0.5°$, regulating from $\theta_0 = 0.08\,\text{rad} \approx 4.6°$ under actuator limits $|u(t)| \leq 20.0\,\text{N}$.

Controller	Settle θ t_s [s]	RMSE θ [rad]	Energy E_u [N^{2}s]	Peak $\lvert u\rvert_{\max}$ [N]	S
LQR (Full State, Clean)	0.942	0.0150	**1.49**	6.97	Idea
LQG (Optimal Filter)	2.070	0.0249	**2.21**	**3.28**	**Optin**
Luenberger + K	0.496	0.0165	**16.20**	**15.10**	Severe N
LQG (Overconfident $V/100$)	0.828	0.0215	2.97	4.80	Noise

Table 5: Benchmark results for Experiment 06 comparing LQG, Luenberger observer, and clean LQR under measurement noise (from `experiments/06_lqg_vs_lqr_measurement_noise/table.md`).

Key Physical and Estimation Insights:

1. **The Role of the Kalman Filter (Noise Rejection over Raw Bandwidth):** The Kalman filter's objective is to prevent sensor noise from corrupting actuator commands. LQG achieves the lowest control energy ($2.21\,\text{N}^2\text{s}$) and lowest peak force ($3.28\,\text{N}$) of any noisy-sensor controller, with a visibly smooth control trace (Figure 4b). It settles in $2.07\,\text{s}$ (vs. $0.94\,\text{s}$ clean LQR) precisely because trading estimator bandwidth for noise rejection is the mathematically optimal compromise.
2. **The Cost of Fast Pole Placement Without a Noise Model:** Fast pole placement (`Luenberger + K`) achieves fast settling ($0.50\,\text{s}$) but directly amplifies raw encoder noise into violent high-frequency actuator chattering, consuming **11× more control energy** ($16.20\,\text{N}^2\text{s}$) and demanding $15.1\,\text{N}$ peak force. Fast observers without noise awareness destroy actuator longevity.
3. **Significance of Covariance Tuning (V):** An overconfident Kalman filter ($V/100$) overtrusts noisy measurements, increasing control effort ($2.97\,\text{N}^2\text{s}$) and peak force ($4.80\,\text{N}$) while increasing vulnerability to disturbance spikes.

4. **Near-Ideal Reconstruction from Partial Sensing:** Reconstructing 4 dynamic states from just 2 noisy encoders gets within 1.7× the clean-LQR angle RMSE while demanding *lower* peak force (3.28 N vs. 6.97 N), confirming that the Separation Principle operates effectively on real nonlinear systems.

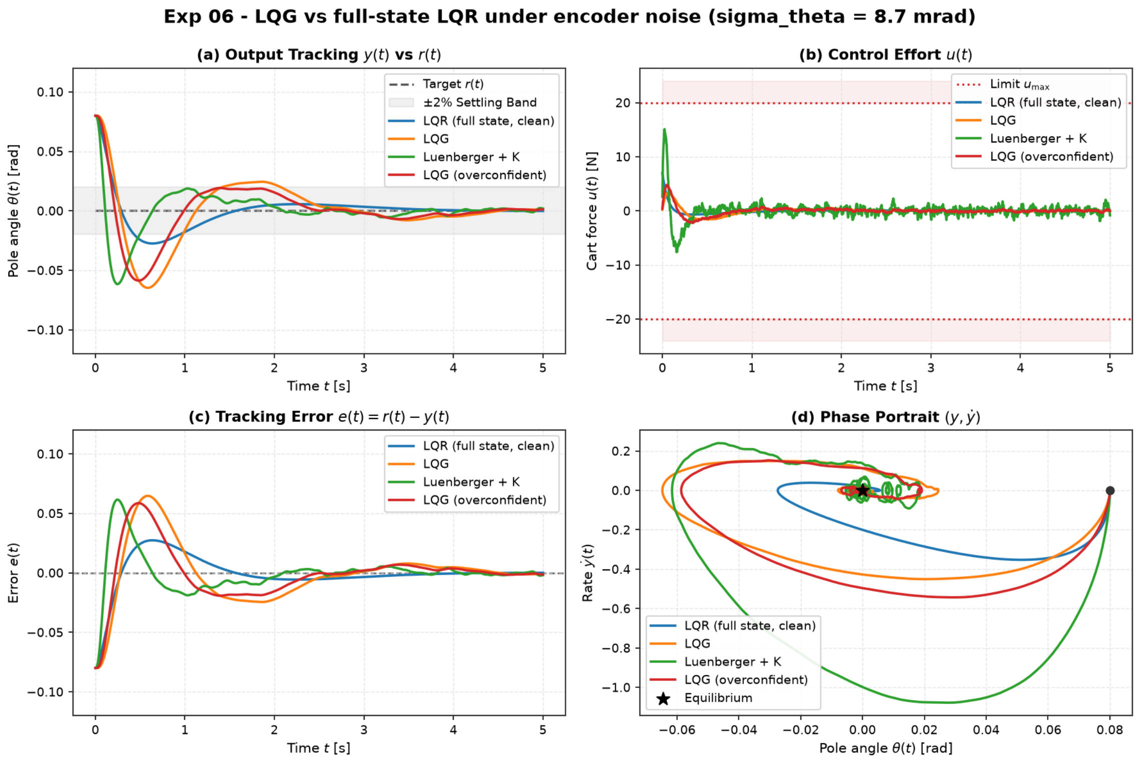

Figure 4: Experiment 06 benchmark comparison under encoder noise ($\sigma_x = 5\,\text{mm}, \sigma_\theta = 0.5°$). (a) Pole angle tracking $\theta(t)$, (b) Control force $u(t)$ contrasting LQG smooth action against Luenberger violent noise chattering, (c) Tracking error, and (d) Phase portrait $(\theta, \dot{\theta})$.

6.5 Nonlinear Basin of Attraction: Cart-Pole LQR Robustness (Experiment 05)

A linear controller $u = -Kx$ designed on the upright linearisation operates within a finite **basin of attraction** on the true nonlinear plant. In Experiment 05, we quantify the boundary of recoverable initial conditions $(\theta_0, \dot{\theta}_0)$ across three reference LQR tunings under hard actuator saturation $|F| \leq 20.0\,\text{N}$.

Tuning	**Effort** R	**Measured** θ_0 **Edge**	**Lyapunov Est.**	**Measured** $\dot{\theta}_0$ **Edge**	**Lyapunov Est.**
Balanced	0.10	**0.83 rad (48°)**	0.80 rad (46°)	**4.4 rad/s**	4.0 rad/s
Aggressive	0.01	**0.92 rad (53°)**	0.80 rad (46°)	**5.3 rad/s**	4.0 rad/s
Soft	1.00	**1.00 rad (57°)**	0.33 rad (19°)	**5.3 rad/s**	1.3 rad/s

Table 6: Measured recoverable boundaries vs. theoretical Lyapunov sub-level sets on nonlinear Cart-Pole.

Key Physical and Algorithmic Findings:

1. **Replication of Literature Envelopes:** The from-scratch CARE solver precisely replicates the canonical Lyapunov envelope (0.83 rad/4.4 rad/s measured vs. 0.80 rad/4.0 rad/s predicted).

2. **Conservatism of Quadratic Lyapunov Sets:** The sub-level set $x_0^\top P x_0 \le c^*$ provides an inner conservative bound. For the soft tuning, the true nonlinear recoverable angle is $\sim 3\times$ larger (1.00 rad vs 0.33 rad) because the quadratic form P is not aligned with the nonlinear invariant manifold.
3. **Actuator Saturation Expands the Basin:** Unconstrained linear feedback over-commands extreme forces at large angles ($\theta > 50°$), leading to rapid numerical and physical divergence. Hard clamping to ± 20 N produces smoother, recoverable trajectories.
4. **Boundary for Nonlinear Control:** Beyond $\theta_0 > 57°$, linear state feedback fails definitively, establishing the explicit operating domain where **nonlinear energy-based swing-up** is required.

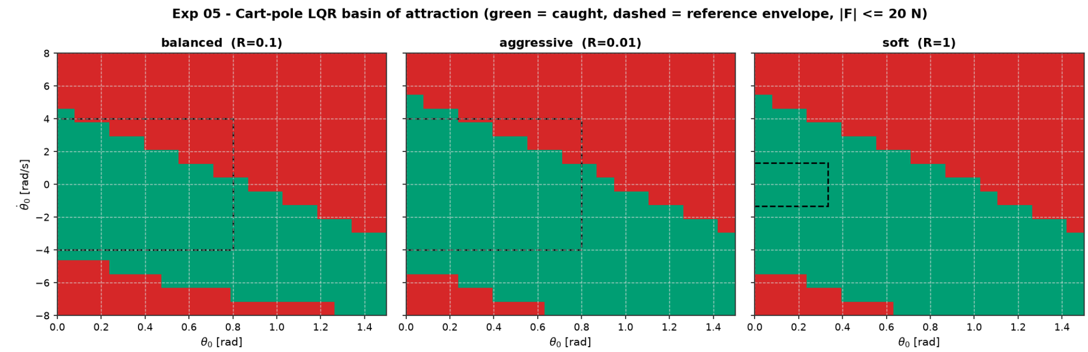

Figure 5: Experiment 05 2D ($\theta_0 \times \dot\theta_0$) basin of attraction maps comparing Balanced, Aggressive, and Soft LQR tunings under ± 20 N actuator limits on nonlinear Cart-Pole.

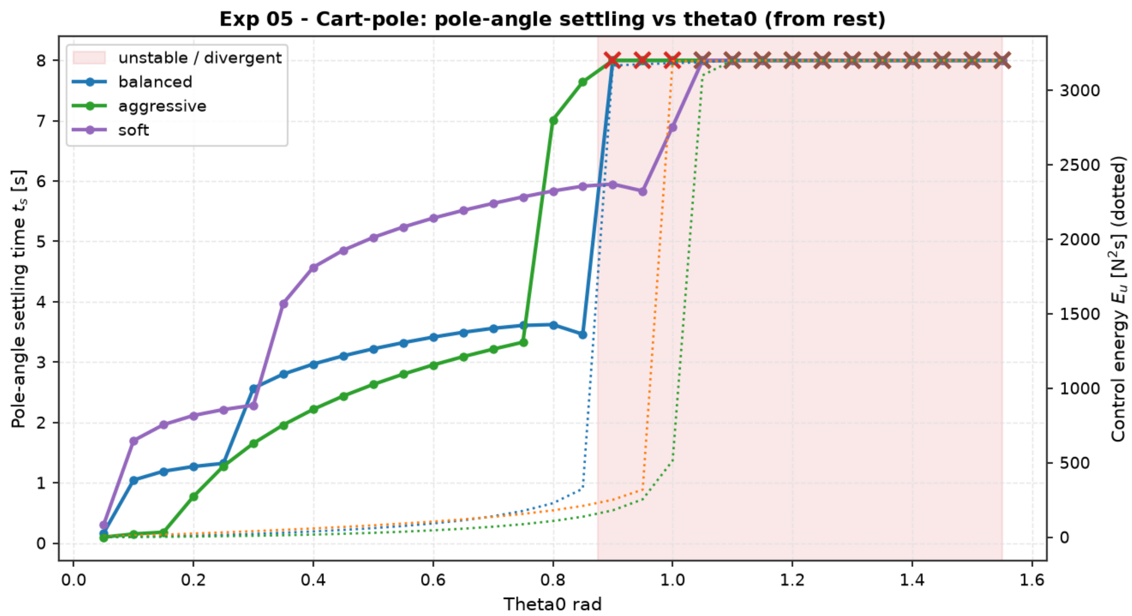

Figure 6: Experiment 05 1D initial angle sweep from rest ($\theta_0 \in [0, 90°]$) illustrating settling time t_s (left) and control energy E_u / peak force $|u|_{\max}$ (right).

7 Project Status and Roadmap

Milestone M3 — State Estimation + LQG — Completed. Library core, four reference dynamical systems, RK4 simulator, PID with anti-windup, StateFeedback (Ackermann), LQR (from-scratch CARE solver), ObserverFeedback, complete state estimation suite (`aimct.estimation`:

Luenberger observers, Filter ARE solver, continuous/discrete Kalman filters, observability matrix), 13 benchmark metrics, multi-controller comparison harness, parameter sweep engine, Experiment 03 (PID unstable stabilization), Experiment 04 (Cart-Pole LQR vs. Pole Placement vs. PID), Experiment 05 (Cart-Pole basin of attraction mapping), Experiment 06 (LQG vs. LQR under measurement noise), CI on Python 3.10–3.12 with 132 passing unit tests, and theory notes for Modules 01–05.

Phase 1 & 2 Priorities (Next).

- **Nonlinear Control & Swing-Up:** Energy-based swing-up controller for the Cart-Pole to bridge the gap from inverted down ($\theta_0 = \pi$) into the linear basin ($\theta_0 \leq 0.83\,\mathrm{rad}$).
- **System Identification:** Data-driven linear state-space fitting for L1 and L2 benchmark systems.
- **Phase 2 (Optimal Control):** Constrained Model Predictive Control (linear MPC with from-scratch QP solver).

Reproduction Commands.

```
git clone https://github.com/zalihthomas-ui/ai-meets-control-theory.git
cd ai-meets-control-theory
pip install -e ".[dev,xcheck]"
pytest -q
python experiments/03_pid_stabilizes_unstable/run.py
python experiments/04_lqr_vs_pole_placement_cartpole/run.py
python experiments/05_cartpole_basin_of_attraction/run.py
python experiments/06_lqg_vs_lqr_measurement_noise/run.py
```